# Auditable Discovery Records:
## Telling Development from Reconstruction in a Dated First-Person Archive

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**Abstract**

Records kept while a theory is being built are evidence about how it was built, and can test its builder's later account. This paper develops a protocol for such tests and applies it to my record of one framework's construction, from timestamped journal entries of 2022 to a preprint independently dated November 2024. The protocol separates what each kind of artifact can establish, codes changes between formulations component by component, and tests later accounts claim by claim. Three transitions are audited: a redesign of the framework's rules, a clarification on four stated interpretive assumptions and a rule of scope, otherwise a contraction or a revision; a change to its third postulate that adds a condition and rejects a too-strong candidate on the page; and a distinction drawn without a name in 2023, renamed and generalized in 2024. Of twenty-three claims selected from my later accounts, the record contradicts six, including three misdated firsts in my own wording. A two-stage account of breakthroughs and a distinction between installed and independently constrained structure are restated as a hypothesis and a coding aid, and the one track scored does not show the first. The paper does not assess whether the framework is true.

## 1 The problem, and a disclosure

The discovery side of science is not unstudied. Its notebooks have been read closely and well: Gruber on Darwin [2], Tweney on Faraday [3], Holmes on Krebs [4], Nersessian on Maxwell [5], with Hanson and the discovery-logic debates collected by Nickles behind them [6, 7]. Reasoning has also been recorded as it happened. Dunbar spent a year in the weekly meetings of molecular-biology and immunology laboratories, studying how researchers used analogies and handled unexpected results [8, 9]. And mathematics now has public records of discovery in progress, written by the participants: the Polymath projects worked in timestamped blog comments, dead ends included, so that, as Gowers and Nielsen put it, "for the first time one can see on full display a complete account of how a serious mathematical result was discovered" [10, 11].

All of these records meet the same question when a later account is set beside them. Published papers present finished structures, and the complaint that they misrepresent the work behind them is old [12]; Holmes, reading Lavoisier's successive drafts, found new ideas arising in the act of writing [13]. Memoirs and retrospectives do the same thing at longer range. When a later

*Email: akb9408@rit.edu, Website: `https://aethus.org`. This paper draws its data from the dated record of one framework's construction, much of it reproduced in the long companion manuscript [1]. It is written in the first person because its data are first-person (§1). Every passage it quotes or tests is frozen, with its location and the SHA-256 hash of its source file, in a supplementary source dossier.

formulation is set beside an earlier one, three things can be going on: the later text renames or clarifies what the earlier one already contained; the content changed; or the later account reads into the earlier text what was not there. This paper asks how a dated record can tell these apart, and answers with a protocol applied to one record.

The record is mine. It documents the construction, between 2022 and 2026, of a framework in the foundations of quantum theory [1, 14]: journal entries timestamped as they were written, two manuscripts of 2023, recorded talks, and a preprint deposited in November 2024 whose version history is kept by the archive rather than by me. A first-person record adds something third-party reconstruction lacks, the builder's own reading of his notes, and that reading can be tested against the notes themselves. It also carries an obvious hazard: the person auditing the record is the person whose story it tells. The design answers the hazard in three ways. Claims about the record are routed to artifacts a reader can inspect, and the passages used are frozen in a supplementary dossier; each code is entered with the interpretive assumptions it needs, so that a reader who disagrees can locate the disagreement; and the audit is run against my own later accounts, several of which it finds contradicted.

The contribution is accordingly narrow. Longitudinal first-person records are not new, nor are graded provenance and the worry about retrospective reconstruction. What this paper offers is their combination: a long first-person record, a separation of the properties each artifact can certify, explicit interpretive rules, developed during this analysis and frozen in this version, and tests of later accounts against earlier artifacts. Against the notebook studies, the record adds its author's own reading of his notes, open to test against them, and, for part of its span, dating by a third party. Against Dunbar's laboratory data, it adds a span of years and the private stages that come before any group meeting. Against Polymath, it adds an individual's pre-public stages, at the price that its private leg is dated by its author rather than by a platform. It has one subject, self-selected, and its codes are so far mine.

Now the disclosure, which the design makes mandatory rather than polite. I hold methodological views that classify my own framework favorably, and the framework was built to satisfy two criteria of adequacy that I held before it existed. This paper neither states nor uses those criteria, which I develop separately. A construction engineered to pass a test contributes nothing to the test's validity, so the framework's scores on those criteria are no evidence for them, and nothing here depends on them. This paper's conclusions concern process, what the record shows about how the framework was built, and a reader who rejects the framework should be able to assess them on the record alone. Whether the framework is true is a question for its own papers, under independent refereeing.

One declaration of genre. The paper is written in the first person because its data are first-person. The method has a small lineage — Einstein's autobiographical notes, Poincaré's and Hadamard's essays on invention, Polanyi's personal knowledge [15, 16, 17] — whose common understanding is that some evidence about invention exists only in the first person.[1] Sections 2–4 say what a discovery record can be evidence for, describe this one, and state the protocol; Section 5 audits three transitions and Section 6 tests later accounts; Sections 7–9 restate two of my earlier theses as a hypothesis and a coding aid and discuss judgment at points of revision; Section 10 gives the record's limits, a statement on AI assistance, and the practices the analysis supports.

[1] Reflexivity is not new to the study of science. The Strong Programme made it a tenet [18], and Ashmore performed it on the page [19], where textual self-reference spiraled toward regress. The self-application here is of a different kind: it ends in comparisons between dated artifacts rather than in further text about itself.

## 2 What a discovery record can be evidence for

Let $D$ be discovery-record evidence, $H_P$ a hypothesis about the process by which a theory arose, and $H_T$ the hypothesis that the theory is true. $D$ bears directly on $H_P$: on what was written and when, on whether idea $A$ preceded idea $B$, on whether a later account matches the contemporaneous notes. Its bearing on $H_T$ is indirect. A record can show that a theory was fitted before a held-out dataset was seen, that an apparent derivation was reverse-engineered from its answer, that independent routes converged on one result, or that reporting was selective. Each is a fact about the process that rationally moves confidence in $H_T$, because each bears on the reliability of the method that produced the theory; the debate over prediction and accommodation turns on exactly such facts [20, 21].

Literal Bayesian screening, $P(H_T \mid D, E, M) = P(H_T \mid E, M)$, holds when $H_T$ is independent of $D$ given $(E, M)$, with $E$ the object-level evidence and $M$ independently validated information about the method's reliability. Where $D$ informs $M$, the record bears on truth through that channel. The shape of the story is another matter. A discovery narrative with a favored shape — an epiphany, a surprise, an early insight later vindicated — does not confirm the theory by having that shape. Whatever confirmatory force the shape has requires independent justification connecting it to a reliable process, and where that justification exists, the weight rests on the process the shape indicates rather than on the shape. Every question this paper puts to its record is a question about $H_P$, which is why the disclosure of Section 1 leaves the paper's conclusions standing, although one process fact the audit turns up bears on truth through the channel just described (Section 5.2).

This position on the old separation of the context of discovery from the context of justification [22, 23] drops the separation for content, since records of discovery are evidence about process, and keeps it for normativity, since a record's shape is no warrant by itself. The position is not new. Hoyningen-Huene, cataloguing the distinction's versions, located its core usefulness in separating descriptive from normative perspectives on science [24, 25]; the essays collected by Schickore and Steinle revisit it from historical and philosophical sides [26]; and Aufrecht has argued that Reichenbach's own version does not map onto Hoyningen-Huene's, so that the distinction has no single clear meaning [27]. The nearest relative of the view that rejecting universal rules of induction need not mean abandoning rational assessment is Norton's material theory, on which there are no universal inductive schemas and the warrant of an induction lies in facts of its domain [28, 29]. Section 9 says what the record adds to that view, and what it does not.

## 3 The record, and what each artifact establishes

The record has a private leg and a public leg. The private leg is a run of journal entries, several timestamped to the minute as they were written; the entries used here are in a public export of the journal in my repository [30], and most are also reproduced in the long companion manuscript [1]. The public leg begins with two manuscripts of 2023 [31, 32], hosted with a verbatim restoration and a separately marked annotation layer of 2026 [33]. It continues with a summary document and recorded talks of October 2023 and after [34, 35], and with a preprint deposited at PhilArchive on November 14, 2024, whose version trail records every later deposit, 519 of them by October 19, 2026 (`https://philarchive.org/versions/BENARA-5`). The record also reaches back into an earlier framework, built between 2018 and 2022, and, at memoir grade, into childhood; neither is audited here.

The artifacts differ in what they can establish, and the differences are easy to blur. Integrity

(these are the bytes that were deposited), existence by a date, publication, the order of events, the presence or absence of words, and an author's later account are separate properties, and no single artifact certifies all of them. Timestamping protocols draw the line exactly: identifying data by a hash is one thing, and establishing that the data existed before a given time is another, which requires a trusted third party or an independently dated anchor [36, 37]. Table 1 separates the properties, with the record's instances, and Table 2 lists the artifacts the audit uses.

Table 1: Kinds of evidence in a discovery record, what each can support, and the record's instances.

| Evidence | What it can support | What it does not establish by itself | Instance in this record |
|---|---|---|---|
| Independently dated deposit | That the deposited content existed by that date | First conception, originality, correctness | Preprint, version 1, PhilArchive, 14 Nov. 2024 |
| Platform-dated upload | That the recording existed by the upload date | When its content was first conceived; a transcript's fidelity | Talks posted October 2023 |
| Cryptographic hash | Identity and integrity of particular bytes | When the bytes were created, or when what they record happened | Hashes in the source dossier; a manifest hashing the math-journal master file, 11 Sept. 2026 |
| Hash anchored in an independently dated record | That matching bytes existed by the anchor's date | That the record is complete | None for the private leg |
| Signed message from a third party (DKIM) | Origin domain and integrity; a date asserted by the sender's systems [38] | What the event meant to its recipient | arXiv account confirmation, 8 May 2023 |
| Local metadata or self-hosted dating | The chronology the author recorded | An independently certified chronology | Journal timestamps; PDF metadata of the 2023 manuscripts |
| Presence or absence of terms | That the terms occur, or do not, in the inspected text | Exposure to, or isolation from, the associated ideas | Term counts in version 1 |
| Retrospective recollection | The author's later account of an episode | A contemporaneous record of the episode | The long manuscript's genealogy; the 2026 annotations; parts of this paper |

The strongest independent anchor is the preprint's version trail. It is an existence-by-date certificate: it shows that the content of version 1 existed by November 14, 2024. Any stronger claim, that the content was mine first or first anywhere, needs separate argument, and this paper does not make one. Before the preprint, the independently dated artifacts are few: the talks posted in October and November 2023, and the signed messages of a correspondence archive, which date events such as the opening of an account rather than the content of the work. The journal entries and the two 2023 manuscripts are dated by their author, and their order is checkable for internal consistency and no more. Some of that checking can be done: the February 2023 entry quotes the January manuscript's three rules verbatim as its "Old Version" (Section 5.1), which is what the January manuscript's priority predicts, and the entry of May 7, 2023 states as a "Postulate" the one-generation condition that the May manuscript proposes and rejects (Section 5.2). A hash fixes bytes, not the dates of what the bytes record: the September 2026 manifest identifies the

Table 2: The artifacts the audit uses. "Self" marks dating under the author's control.

| Artifact | Date | Dating basis | Used for |
|---|---|---|---|
| Journal entry, "Explanation For Double Slit Experiment with Centric Unfolding" | 21 Aug. 2022, 10:27 AM | Self (journal) | "Compress" preferred to "collapse" |
| Journal entry, "Double Slit Experiment With Human Subjects" | 22 Aug. 2022, 1:23 and 1:42 PM | Self (journal) | The unknown set dropped; the Anti-Ethus defined; the question of future states |
| Journal entry, "Rules for Superposition" | 23 Oct. 2022, 10:37 AM | Self (journal) | Rules #1–#5; the worry about communicating with the future |
| Journal entry, "Severed-Ethus" | 10 Jan. 2023, 7:06 PM | Self (journal) | A construct not carried forward |
| Manuscript, *Centric Unfolding* | "January 2023" on its face; 28 Jan., 14:51:29 UTC by PDF metadata | Self (hosted) | Three rules; child and parent defined; the paradox of footnote 6 |
| Journal entry, "Rephrasing the Three Laws of the Ethus" | 8 Feb. 2023, 4:42 PM | Self (journal) | Old and new laws, with reasons |
| Journal entry, "Revisiting the Measurement Problem" | 7 May 2023, 10:54 PM | Self (journal) | A one-generation candidate stated as a "Postulate" |
| Account-confirmation message, arXiv | 8 May 2023 | Third party (DKIM) | An account event |
| Journal entry, "Plan for arXiv Publish" | 9 May 2023, 6:24 PM | Self (journal) | The account event tied to a judgment that the work was done |
| Manuscript, *Solving the Measurement Problem* | "May 2023" on its face; 23 May, 17:10:04 UTC by PDF metadata | Self (hosted) | Three postulates; a rejected candidate; two superpositions |
| Summary document of the rules (Google Docs export) | 20 Oct. 2023 by the site's listing | Self (hosted export) | The superpositions unnamed; the "neither" case kept |
| Recorded talk, "Did I Just Solve the Measurement Problem?" | 19–20 Oct. 2023 (the displayed day varies with time zone) | Platform | The postulates in their May wording; the superpositions unnamed |
| Draft video series | May–Aug. 2024 | Self (listed on the author's site; platform record not linked) | "Blank"; the names of the two superpositions |
| Preprint, version 1 | 14 Nov. 2024, 23:40:09 GMT | Archive | Existence by date; term counts |
| Long manuscript, version 519; annotation layer; earlier draft of this paper | 2026 | Archive; self; self | The later accounts tested in Section 6 |

math-journal master file from the date its hash was recorded, and the entries inside the file remain dated by their author.

Four inferences from such artifacts are easily overdrawn. First, across version 1 of the preprint the phrase "possible world(s)" occurs, by my count, fifty-six times, while "counterfactual," "Lewis," "Stalnaker," and "perturbation" do not occur (the dossier does not hold version 1, and lists these counts as unverified). That establishes what the text contains. It is compatible both with isolation from the modal literature and with exposure that went uncited, so it cannot establish isolation by itself. Second, the January 2023 manuscript has no citations; that establishes what it cites, not what its author had read. Third, a DKIM-signed message from arXiv dates the opening of an account on May 8, 2023. A journal entry dated the next day ties the event to a judgment that the work was done: "yesterday, right after, (at least I hope), officially cracking the measurement problem, I made an arXiv account." The entry is self-dated, and the message dates only the event. Fourth, a later name for an earlier construction does not establish that the earlier and later contents are the same. That has to be shown, which Section 5.3 attempts for one pair of names.

Two derived artifacts need grading of their own. The talks are quoted from transcripts made by speech recognition and repaired by a logged rule list restricted to the framework's coined terms. The transcripts are a reading aid, and a quotation that matters should be checked against the audio at the timestamp given; below, only phrases in standard English are quoted from them, no coined term is counted from them, and the one exception, the first recorded use of two coined names, is flagged where it occurs. The 2023 manuscripts are quoted from the deposited PDFs, against which every quotation here has been checked; the verbatim restoration reproduces them, slips included, and keeps the 2026 annotations visibly apart from the text. The passages this paper quotes or tests are frozen in the supplementary dossier, each with its artifact, version, location, and the SHA-256 hash of its source file, so that a checker need not rely on any later edition.

## 4 An audit protocol

A transition is an ordered pair of formulations of one construct, each tied to an artifact in Table 2. Constructs are identified by function, not by name: the third rule of January 2023 and the third postulate of May 2023 are formulations of one construct because each is the framework's condition on which states can be realized. By the same criterion, a statement of how a state is updated belongs to a different construct from a statement of which states are real, in superposition, or valid. Each formulation is divided into components, claims that can be located in its text, and the coding proceeds component by component. A transition's code is derived from its components' codes, and a later account is tested claim by claim. Table 3 defines the codes and verdicts.

Five rules govern the coding.

(i) *Sense by use.* A term's content is fixed by how the artifact uses it, not by its textbook sense and not by the sense a later text gives it. Table 4 records the senses used here, and Section 8.2 explains why this rule carries weight in this record.

(ii) *Two-way preservation.* A transition is a clarification only if every component of each formulation is preserved in the other. One-way entailment is not enough: a later formulation entailed by the earlier may have dropped some of its content, as $A$ is entailed by $A \wedge B$.

(iii) *Stated interpretations.* Each component code is entered with the interpretive assumptions it needs. A code that holds only on an assumption is conditional on it, and the transition is reported with the code it receives if the assumption is denied.

Table 3: Codes for components and transitions, and verdicts for later accounts.

| | |
|---|---|
| *Component codes, for each component of either formulation* | |
| Preserved | Stated or entailed in the other formulation, under the interpretation entered with the code |
| Renamed | Preserved under a new term whose extension, read by use (rule i), is unchanged |
| Added | A component of the later formulation, neither stated nor entailed in the earlier |
| Dropped | A component of the earlier formulation, neither stated nor entailed in the later |
| Indeterminate | The texts do not settle which of the codes above applies; the rival codes are named |
| *Transition codes, derived from the component codes* | |
| Renaming | Every component preserved, some under new terms, nothing else changed |
| Clarification | Every component of each formulation preserved in the other; wording, explicitness, or organization changed |
| Extension | Some component added, none dropped |
| Contraction | Some component dropped, none added |
| Revision | Components both added and dropped |
| Mixed | Parts of the transition, such as different cases, receive different codes; each part is reported with its own |
| Abandonment | A construct dropped without successor; where its work is absorbed, if anywhere, is recorded |
| *Verdicts, for each atomic claim of a later account* | |
| Supported | An earlier artifact states or entails the claim |
| Contradicted | An earlier artifact states or entails its negation |
| Compatible, not discriminating | The earlier artifacts fit the claim and a named rival equally |
| Open | The record does not bear on the claim, or bears on it only through an unsettled code |

A code that holds only on an interpretive assumption is conditional on it. Each verdict carries the grade of its best evidence: independent, when a third party dates it, or self-dated.

(iv) *Paired readings.* Each code is entered with the strongest competing reading, stated as the denial of a named assumption or as a rival code for a named component, so that a coder who disagrees can say where. A transition whose parts receive different determinate codes is mixed; a component whose code the texts do not settle is indeterminate. The first is a finding about the transition, the second about the evidence.

(v) *Outcome-blind coding, and direction of test.* A transition is coded from its two artifacts before consulting whether the construct survived; survival is recorded separately. A later account is split into atomic claims, each asserting one thing, and each claim is tested only against artifacts dated before the account.

The codes and rules were developed during this retrospective analysis, by the person whose record it is, and are frozen in this version. They were not preregistered; their place before the transitions they are applied to is an order of exposition. The codes in Sections 5 and 6 cannot meet the first clause of rule (v), since I chose the transitions knowing their outcomes, and they are offered as a first coding for an independent coder to test. A coder with no stake in the framework, working from the frozen passages with Tables 3 and 4 and the coding sheet of Appendix A, can recode the same transitions blind to my codes. Such a recoding tests the reliability of the codes, not the hypotheses of Sections 7 and 8, which need records other than mine.

# 5 Three transitions

The three transitions were chosen, with their outcomes known, to exercise the codes: a redesign whose code turns on stated interpretive assumptions, an extension with a rejected candidate on the page, and a naming whose code turns on one component the texts leave undetermined. All three concern the framework's postulates. The first two fall between January and May 2023, so their order rests on the private leg; the first independent anchor for any of their content is the October 2023 talk. Table 4 glosses the framework's terms as the audited versions use them, so that the early texts are not read in the mature framework's senses.

## 5.1 The rules redesigned: a conditional clarification

The January 2023 manuscript was the first manuscript to assemble the framework's rules as three numbered rules, the first two restating rules of a five-rule list in the October 2022 math journal ("Something can only be real relative to an Ethus"; "Any Attribute of an Anti-Ethus will be in superposition relative to its Ethus"):

> 1. Anything may only be real relative to an Ethus. 2. Any attribute of an Anti-Ethus will be in superposition relative to its Ethus. 3. An Ethus is invalid if a contradiction is triggered anywhere in the superposition of its parent Ethuses.

Eleven days later by the file's metadata, on February 8, a journal entry boxed these three sentences as the "Old Version" and replaced them:

> 1. Anything may only be real relative to an Ethus. 2. If an Ethus is valid, then any given property will be in superposition relative to it if and only if it is statistically independent to it. 3. Any child of an invalid Ethus will also be invalid.

The entry, titled as a rephrasing, says it was prompted by "a little bit of a blend between the old second law and third law's descriptive information," and gives two reasons. The old third law reaches "invalid parents based on the given Ethus working 'backwards' in the tree," which "makes

Table 4: The framework's terms as used in the versions audited.

| Term | In use | Sense in use, with the passage that fixes it |
|---|---|---|
| Ethus | Aug. 2022 – Feb. 2023 | A state of knowledge: "Let this set of knowledge be called an Ethus" (Jan. 2023); "the abstract set of knowns something may hold on the world" (journal, 10 Jan. 2023) |
| Aethus | May 2023 on | The same construct renamed: "Let this reality-perceiving entity be referred to as an 'Aethus' " (May 2023) |
| Anti-Ethus | Aug. 2022 – Jan. 2023 | "the set of all possible things which are Rebetary, (Relatively Arbitrary / statistically independent), to the Ethus" (Aug. 2022); it "contains all the properties in reality which would have zero net effect on the Ethus if they were altered" (Jan. 2023). Retired from the laws in Feb. 2023 |
| Statistically independent | Aug. 2022 – July 2024 | Glossed as "Relatively Arbitrary" (Aug. 2022); a property only partly correlated with the Ethus "may simply be subdivided into an Ethus component and an Anti-Ethus component" (Jan. 2023). Whether the term keeps its textbook sense is open (Section 6) |
| Superposition | 2022 on | Holding every admissible value at once, skewed only as the state of knowledge directs: an Anti-Ethus variable "exists in all possible states at once," and the superposition of the Anti-Ethus "will never arbitrarily skew one way or another without the direct order of the Ethus to do so" (Jan. 2023); an independent attribute "will occupy all possible states at once that it may occupy without influencing the Aethus" (May 2023) |
| Contradiction | 2022 on | Used from 2022 ("No contradictions allowed!", Oct. 2022); defined in January 2023 as "an Attribute in the Ethus holding two possible values instead of just one" |
| Invalid | Jan. 2023 on | A state that contains or triggers a contradiction, and so cannot be realized: an invalid Aethus "will forever stay on paper, and will never exist in our physical universe" (May 2023) |
| Child, parent | Jan. 2023 on | "a child Ethus as meaning an Ethus with added information from another Ethus, then a parent Ethus would be the product of removing information from an Ethus" (Jan. 2023) |
| Compression | Aug. 2022 on | Update without collapse: "I like the word 'compress' better" (Aug. 2022); "A superposition may never truly collapse – rather it can only compress" (Jan. 2023) |
| Rule, law, postulate | Oct. 2022 on | The framework's basic statements: rules (Oct. 2022); rules and "basic laws," with its principles called "postulates" in a footnote (Jan. 2023); laws (Feb. 2023); postulates (May 2023 on) |
| Blank | 2024 on | Distinguished from independence: "not statistically independent ... but it's still blank" (draft video, June 2024) |
| Agreeing, disagreeing superposition | 2024 on | Names for the two superpositions of Section 5.3, with a definition for any attribute (draft video, July 2024) |

for good practical application, but isn't very law-like"; and the old formulation "doesn't work to hint at the clear correlation between the contradiction and the clash of Anti-Ethus superposition," which the new version "fixes quite nicely." In the new version each concept is "mutually exclusive," in the entry's phrase, from the last, and the entry singles out the new second law's contrapositive, which yields a test for invalid states, as "a nice touch."

Whether the redesign added or dropped content turns on entailment, and Table 5 sets out the comparison. It lists each component of the February laws with the January passage that supplies it, then each January component with its place in the February laws, and enters the interpretive assumption, if any, on which each code depends. Each direction of the second law is entered together with its contrapositive, which is one of the two cases of invalidity the entry derives from the law. Rows that carry January content through a term, such as "superposition," "child," and "statistically independent," need no assumption beyond rule (i): the February entry gives these terms no new sense, so it uses them in January's. Four assumptions do the work.

**I1.** A contradiction triggered in a parent's superposition makes the parent invalid. January states as much ("if an Ethus with removed information from another causes a contradiction, then both Ethuses are invalid") and asserts the resulting equivalence itself, calling the child-directed form "another way of putting it."

**I2.** To be in superposition is to hold more than one value. January says of an Anti-Ethus variable that "it exists in all possible states at once," and calls an attribute of the Ethus "holding two possible values instead of just one" a contradiction, "which unlike the Anti-Ethus, is strictly forbidden."

**I3.** January's rules state what holds of an Ethus that can be realized, so that an Ethus that breaks one is invalid rather than a counterexample to it. January treats a breach of rule 3 this way ("rule number three would be broken if such an Ethus was permitted, and so, blatantly speaking, it is not"), and a breach of the Anti-Ethus's defining independence too: "Even though an Anti-Ethus, in its definition, is assumed to be statistically independent of the Ethus, it still holds that any Anti-Ethus which clashes somewhere with its Ethus, hypothetical or practical, would be collectively invalid in the real sense" (footnote 2). No passage says it of rule 2, so I3 extends the pattern.

**I4.** Footnote 3's check states a necessary condition for validity, not a sufficient one. Its "Therefore" follows from the claim that "an invalid Ethus, in being functionally contradictory, may only beget invalid children Ethuses," which yields only that an Ethus with a contradicting ancestor is invalid.

Table 5: Audit of the first transition: the components of each formulation, sought in the other.

| Component | Where the other formulation supplies it | Needs | Code |
|---|---|---|---|
| *Construct*: the framework's conditions on states: which are real, which properties are in superposition, which states are valid. *Earlier*: January rules 1–3, read with the passages of §§3–4 and footnotes 2–3 that state or gloss their content ("January 2023"; PDF metadata 28 Jan., self). *Later*: February laws 1–3 (8 Feb. 2023, 4:42 PM, self; the entry quotes the January rules verbatim). | | | |
| *February components, sought in the January text* | | | |
| Law 1 | Rule 1, verbatim | — | Preserved |

| Component | Where the other formulation supplies it | Needs | Code |
|---|---|---|---|
| Law 2, forward, with its contrapositive: an independent property of a valid Ethus is in superposition; an Ethus with an independent property out of superposition is invalid | Rule 2, with §3: the Anti-Ethus "contains all the properties in reality which would have zero net effect on the Ethus if they were altered. In other words, the Anti-Ethus is statistically independent to the Ethus." Rule 2 states this of every Ethus, which entails it of valid ones | — | Preserved |
| Law 2, converse, with its contrapositive: a property in superposition relative to a valid Ethus is independent of it; an Ethus with a property in superposition but not independent is invalid | §4: an attribute of the Ethus holding more than one value is "strictly forbidden," and "Either way, the Ethus in question is invalid"; §3: a partly correlated property is "subdivided into an Ethus component and an Anti-Ethus component." For a clash with the Ethus, footnotes 2–3 supply the case without I2 | I2 | Preserved on I2 |
| Law 3 | §4: "Another way of putting it is that every child of an invalid Ethus is also invalid" | — | Preserved |
| *January components, sought in the February laws* | | | |
| Rule 1 | Law 1, verbatim | — | Preserved |
| Rule 2, stated without condition | Law 2, forward, which states it of valid Ethuses; the Anti-Ethus's content carried by "statistically independent" | I3 | Preserved on I3; dropped if I3 is denied |
| Rule 3, parent-directed | Law 3 | I1 | Preserved on I1 |
| §4: "no contradictions are permitted," a contradiction being an attribute of the Ethus holding two values | Law 2, converse, in its contrapositive, which makes such an Ethus invalid | I2 | Preserved on I2 |
| §3: the superposition "will never arbitrarily skew one way or another without the direct order of the Ethus to do so" | Carried by "superposition," which the laws use in its January sense (rule i; Table 4) | — | Preserved |
| §3: a partly correlated property is "subdivided into an Ethus component and an Anti-Ethus component" | Carried by "statistically independent" in its January sense (rule i); law 2 then places each component | — | Preserved |

| Component | Where the other formulation supplies it | Needs | Code |
|---|---|---|---|
| Footnote 2: an Anti-Ethus that clashes with its Ethus is invalid | Law 2, converse, in its contrapositive: footnote 3 reads the clash as a failure of independence, "defeating its definition of independence" | — | Preserved |
| Footnote 3: to check an Ethus for validity, "one must prove that every possible parent/ancestral Ethus to it will never trigger a contradiction" | Law 3, applied along the ancestry, which gives the check's necessity | I1, I4 | Preserved on I1 and I4; its sufficiency dropped if I4 is denied |
| Child and parent, as defined in §4 | "Child" used in law 3, "parent" as its converse; neither redefined | — | Preserved |
| §4: reality avoids a contradiction "adding to the Ethus accordingly" | Not stated or entailed in the laws | — | Outside the construct: the update rule (see text); dropped if counted |

*Competing reading*: denies I3. January's rule 2 holds of every Ethus, so an independent property is never out of superposition; February states the law of valid Ethuses only and makes its breach a test of invalidity, which drops the unconditional rule. *Code*: clarification on I1–I4. Denying I1, I3, or I4, or counting the update rule, drops a component (contraction); denying I2 also adds the converse (revision). The Anti-Ethus coded abandonment, its work absorbed into law 2. *Survival*: law 3 reappears in May 2023 as a derived consequence (Section 5.2).

On I1–I4, every component of each formulation is stated or entailed in the other, and the transition is a clarification: what changed was which sentence held which content. The child-directed form moved from a gloss into a law, the Anti-Ethus left the laws with its content carried by the biconditional, and the validity condition moved into the second law. The entry's second reason describes a change of this kind, in what the laws "hint at." One January statement falls outside the comparison. The manuscript's section on contradiction says that if a contradiction occurs anywhere in the superposition, "reality will take the simplest possible path to avoid that, adding to the Ethus accordingly." That is a statement of how a state is updated, not of which states are real, superposed, or valid; it continues the October 2022 rule #5, is absent from the February laws, and returns in May (Section 5.2). Counted as a component of the rules, as the competing reading of Section 5.2 counts it, it would be dropped, and no reading would yield a clarification.

Rule (iii) asks for the code each denial yields. Denying I1 leaves rule 3 and the ancestral check of footnote 3 without a counterpart in the laws, a contraction. Denying I3 makes January's rule 2 a law of every Ethus, stronger than the February law, which states it of valid Ethuses only, so the unconditional rule is dropped and the transition is again a contraction. Denying I4 makes footnote 3's check sufficient as well as necessary, as the January derivation of the detector-off result uses it: "we may remove attributes at will from our Ethus as the observer, and never once will a contradiction occur. Therefore, calling the electrons in a superposition long after the experiment is over is totally valid." The February laws state no sufficient condition for validity, so the sufficiency

is dropped, a contraction once more. Denying I2 adds the converse direction of law 2 and leaves January's prohibition of contradiction without a counterpart in the laws, a revision. I3 and I4 are the most contestable: the first extends to rule 2 a pattern January shows elsewhere, and the second reads footnote 3 more narrowly than January uses it. Later evidence does not settle I2: the May manuscript introduces a "converse" of the rule as a further postulate but states the same direction both times ("any concept in superposition relative to an Aethus ... is statistically independent to it"; "if an item is in superposition with respect to an Aethus, then it must be statistically independent to it"). My 2026 account, which calls February 8 "the first axiomatic designation" of the framework, is tested in Section 6.

### 5.2 The third postulate: an extension

The February law said that invalidity passes from a state to its children. It gave no condition under which a state is protected from a contradiction lying in its future, and beyond the cases the second law's contrapositive catches, it said nothing about which states are invalid in the first place. The May 2023 manuscript builds such a condition in four steps, all on the page.

It first proposes a one-generation candidate and rejects it: "We may think to perhaps define this as follows: 'If an Aethus is valid, then no child Aethus of it may contradict any of its properties.' This is close to the final realization, but still not quite accurate." The same one-generation condition, in substance, stands as a "Postulate" in a journal entry of May 7, 2023, 10:54 PM: "no child Aethus of mine contradicts that item in any way." The candidate is too strong, the manuscript explains, because a valid state can have contradicting children, since "simply because there are other options of child Aethae to avoid that signifies that all hope was not yet lost." It then states the condition for a state that is, in its phrase, "necessarily doomed": "For some central Aethus, if for every one of its child Aethae there exists a further child Aethus that contradicts any of its properties, then said central Aethus is invalid." It takes the contrapositive as the postulate:

> Postulate 3 (The Postulate of Aethic Certainty) If some Aethus is valid, then there exists a child Aethus to it for which every one of its own child Aethae strictly do not contradict any of the properties of said Aethus.

Finally it derives the earlier forms as consequences: "every invalid Aethus must have all invalid child Aethae," and "your Aethus is invalid if any one of your parent Aethae is invalid."

Coded by components against the February laws, the transition adds two components and drops none (Table 6). The existential–universal condition over two generations, with the criterion of invalidity that is its contrapositive, is neither stated nor entailed in those laws. Nor is the rule that a state no test shows invalid is valid ("If it also fails to prove your Aethus is invalid, then it must be valid"). On I4 the January derivation of the detector-off result relied on that rule unstated; denying I4, the rule restores the sufficiency of footnote 3's check, which the February laws dropped (Section 5.1). The February third law is preserved as a derived consequence, and so, on I1, is the January rule; the second law reappears as the second postulate; "Ethus" becomes "Aethus" and "laws" become "postulates," renamings by the sense-by-use rule (Table 4). The code is extension, and the candidate is coded abandonment, its rejection stated in the text that proposes it. The two-generation condition has consequences the earlier forms lack: the resolution of the January paradox (Section 5.3), and a derivation that the records of different observers cannot conflict, which the January manuscript does not contain. The detector-on and detector-off predictions are not among them, since the January manuscript already derives both, the first from its third rule and the second from its second and third with the uncertainty principle and the check of footnote 3.

The strongest competing reading denies the construct criterion of Section 4 and counts January's update rule as part of the condition. In January, reality responds to a contradiction anywhere in the superposition by "adding to the Ethus accordingly," and the May text motivates the postulate by the same mechanism ("the universe diverts you away from such an invalid child Aethus by adding a property to your Aethus"). Since the addition removes the contradiction from the whole superposition, January's rule arguably has the existential–universal structure in dynamic form, which May restored. Relative to the February laws the condition is still added, so this transition's code stands, but the first transition becomes a contraction (Section 5.1). On either reading the criterion of a doomed state, for which no addition can avoid a contradiction, is new in May. A weaker reading, January's check over "every possible parent/ancestral Ethus" turned toward children, fails on the quantifiers: a universal check over ancestors is not an existential–universal condition over descendants.

The dating is weaker than the content. The manuscript dates the two-generation form to May 2023, its deposited file produced on May 23 by its metadata. May 8 is an inference from the journal: the entry of May 7 still states the candidate and argues the detector case with it, and the entry of May 9 places "officially cracking the measurement problem" on the previous day, the day of the account event the arXiv message dates (Section 3). Neither entry records the form reached, so the cracking of May 8 may have completed the May 7 argument, with the two-generation form arriving by May 23. The form is independently attested, in the May wording, in the talk posted in October 2023 (43:26), and by my reading in version 1 of the preprint; the independently supported claim is existence by October 2023.

One process fact here bears on more than process. In the October 2023 talk the speaker says he "figured I'd start with some trial and error for the third postulate, see how they perform, and then update it to make it better" (40:17), with the double-slit cases as the test. The postulate was fitted to those cases: it accommodates the detector results rather than predicting them, and by the channel of Section 2 those results provide no independent test of it. Accommodation is not thereby without evidential value; Hitchcock and Sober locate its characteristic risk in overfitting [21], and the place to assess that risk here is the consequence the postulate adds beyond the fitted cases, the agreement of observers' records (Section 8.1).

### 5.3 Two superpositions: a naming and a generalization

The January manuscript runs the double-slit experiment with people walking through two doors on a soccer field, and ends its account with a paradox it cannot resolve. The observer at the back wall knows that each participant went through one door, but not which; "supposing that the observer is unsure of the door the participant took seems to imply that such information would be in superposition by rule number two," and superposition, as the manuscript then understood it, meant going through both doors, which the third rule forbids. The footnote tries two repairs, judges that "[b]oth of these fixes feel somewhat unsatisfactory however for their lack of simplistic clarity," and ends: "Let the readers be the judge of that."

The May manuscript draws the distinction that removes the paradox, without naming it. With the detector running, the third postulate eliminates the state in which the particle went through both slits, and what stays in superposition is that "the particle went through one slit but not the other in both ways." The text anticipates the objection: "It may seem counterintuitive how this is different from the general going through both slits at once, but to Aethic logic it is certainly distinct – for instance, it does imply that one slit is not being passed through no matter what." The October 2023 talk restates the distinction, still unnamed ("A and not B plus B and not A

Table 6: Audit of the second transition: the third postulate.

| | |
|---|---|
| Construct | The framework's condition on which states can be realized |
| Earlier formulation | February law 3: "Any child of an invalid Ethus will also be invalid," with law 2 (8 Feb. 2023, self) |
| Rejected candidate | "If an Aethus is valid, then no child Aethus of it may contradict any of its properties," rejected as too strong on the page that proposes it; stated, in substance, as a postulate in the journal, 7 May 2023, 10:54 PM (self) |
| Later formulation | Postulate 3, in the May wording (quoted above), with the rule of validity by default from the May text's detector-off analysis |
| Date and basis | "May 2023"; produced 23 May by PDF metadata; 8 May by inference from the journal entries of 7 and 9 May (self). Independently: the talk of October 2023 (43:26); preprint version 1, 14 Nov. 2024, by my reading |
| Added | The $\exists\forall$ condition over two generations, and the criterion of invalidity that is its contrapositive; validity by default, relied on unstated in January on I4 and otherwise the sufficiency of footnote 3's check restored |
| Preserved | Law 3 and the ancestral direction (the January rule on I1), as derived consequences; law 2, as the second postulate |
| Renamed | "Ethus" as "Aethus"; "laws" as "postulates" |
| Dropped | None |
| Competing reading | Denies the construct criterion of Section 4: January's update rule arguably has the $\exists\forall$ structure in dynamic form, restored in May; relative to February the condition is still added, and the first transition becomes a contraction. The criterion of a doomed state is new on either reading. The ancestral check turned toward children fails on the quantifiers |
| Code | Extension; the candidate coded abandonment, its rejection stated in the same text |
| Survival | The May wording recurs in the October 2023 talk and, by my reading, version 1; the 2026 statement, over "proper" children, keeps the quantifier structure (not coded here) |

does not equal A and B, and no nots," 46:42), and a summary of the same week records the want of a name: "A clear distinction is now needed between different types of superpositions." The names arrive in 2024. In a draft video listed on my site under July 13, 2024, *Solution to the Measurement Problem*, Part 1, the two cases are introduced as "what I call a disagreeing superposition" and "what I call an agreeing superposition" (8:21; since these are coined terms, this quotation needs checking against the audio), and, by my reading, version 1 of the preprint, dated by the archive four months later, carries both names. I date the naming itself to the turn of 2024, from recollection.

Is the 2024 name a renaming of the 2023 distinction? Component by component, partly (Table 7). On the double-slit case the names attach to the cases the May manuscript distinguished: the disagreeing superposition of the two slits is its "one slit but not the other in both ways," and the agreeing one is its "both slits at once." Those components are renamed, and the insistence that the two differ is preserved. The general definition is added. The 2024 drafts define a disagreeing superposition for any attribute, one alternative at a time switched on and every other switched off, the resulting cases superposed, and an agreeing superposition as the attribute's alternatives superposed together; the 2023 texts draw the distinction only for the two slits, though the May manuscript's list of four cases for the detector-off run, "both slits at once, the first slit but not the second, the second but not the first, or neither," points toward the general construction.

One component is indeterminate: the case in which the particle passes through neither slit. The May manuscript lists it among the detector-off possibilities, and its detector-on analysis reads both ways. Its statement that all that is left is for "the particle to either go through one slit and not the other in either way" excludes it; "just knowing that it did not go through (at least) one of them" and "given that at least one slit was not traversed" leave room for it. The October 2023 talk counts it among the detector-on alternatives ("and I guess neither," 46:42), the summary of the same week keeps it in both runs, and so does Part 2 of the July 13, 2024 draft ("no slits at all, the first but not the second, or the second but not the first," 9:05, detector on; "particle could just miss both slits," 16:40, detector off). Yet both October 2023 sources also restate the detector-on result in May's two-case wording ("one slit but not the other in both ways," 48:26; "to one but not the other in both ways," summary). Whether the May verdicts excluded the case or left it standing, the texts do not settle. On the first reading the later verdicts add a component; on the second they state one that May left implicit, and the October restatements, which keep the case while using May's wording, fit the second. On the second reading the transition is mixed, a renaming on the shared case and an extension in the general definition; on the first it is an extension throughout. Which it is turns on the indeterminate component, a finding about the evidence rather than about the transition.

The competing reading is that the May distinction was posited to absorb the January paradox, with the general typology read back into it later. The May text answers part of that reading, since there the distinction falls out of the second and third postulates applied to the detector case; Section 8.1 states the criterion on which that answer turns. Two later descriptions of this transition are among the claims tested in Section 6.

**The second column.** Beyond the three transitions, the record keeps what did not survive, and the protocol codes it the same way. In August 2022 the earliest design's "unknown set" was dropped as "kind of an overly vague concept" in favor of the Anti-Ethus, which left the laws in February 2023 with its work absorbed. The October 2022 rule that a contradicting superposition's "defining Attribute ... joins its respective Ethus" left the numbered rules in January, where invalid states took its place, but not the record: January's prose keeps its mechanism, the February laws

Table 7: Audit of the third transition: two superpositions.

| | |
|---|---|
| Construct | The distinction between two kinds of superposition |
| Problem stated | January footnote 6 (quoted above); "January 2023" (self) |
| Earlier formulation | "one slit but not the other in both ways" against "both slits at once"; "to Aethic logic it is certainly distinct" (May 2023, self). Restated, unnamed, in the October 2023 talk (46:42; platform) and summary (self) |
| Later formulation | "Disagreeing" and "agreeing" superposition, defined for any attribute |
| Date and basis | Draft video *Solution to the Measurement Problem*, Part 1, listed under 13 July 2024 (self; the coined terms to be checked against the audio); preprint version 1, 14 Nov. 2024 (archive); the naming at the turn of 2024 by recollection |
| Renamed | The two double-slit superpositions, under the new names |
| Preserved | Their difference, and its criterion: "one slit is not being passed through no matter what" |
| Added | A definition for attributes with any number of alternatives |
| Indeterminate | The "neither" case: May's detector-on analysis reads both ways; kept in the October 2023 and July 2024 analyses, which also restate May's two-case result. Rival codes: added or preserved |
| Competing reading | A case-specific device installed against the January paradox and generalized in retrospect. Partly answered by the May derivation from the second and third postulates (Section 8.1) |
| Code | Mixed (renaming on the shared case, extension in the general definition) if May left "neither" implicit; extension if the later texts add it |

omit it, and May uses it again (Section 5.2). A "Severed-Ethus," introduced on January 10, 2023 as "the set of knowns which are permanently in the Anti-Ethus," appears in neither manuscript. The January footnote's two repairs were rejected on the page. "Anti-Time," introduced in a journal entry of February 19, 2023 and named in the May manuscript, appears nowhere after, and the May manuscript's presentation of the first postulate as a generalization of QBism was dropped by later versions. Later still, in February 2026, the reading of quantum uncertainty as a force that conceals which-way records, shared by the 2022 and 2023 texts, was replaced by a reading on which nothing conceals anything [1]. My own later account calls that a change in what is fed into the postulates rather than in the postulates; the protocol would code it as a revision of the framework's physical inputs, and the postulates' own 2026 restatements separately.

## 6 Testing the later accounts

The record's later accounts are of three kinds: the narrative and historian's notes of the long manuscript [1], the 2026 annotation layer of the restoration [33], and an earlier draft of this paper [39]. Each is cited in a fixed version, which the caption of Table 8 names; the dossier gives each passage's location in that version and the hash of its source file.

I selected the claims after coding the transitions. Eligible statements are those in the three sources that date the audited constructs, or describe at a date the artifacts in which they appear; where one sentence makes several such statements, each is entered as a separate claim. Among eligible statements the choice was mine and followed no rule; the tested claims are chiefly firsts, dates, and readings of early entries, one occurrence of each. Eligible statements remain untested, among them statements that place the distinction of Section 5.3, or steps toward it, in 2022 and January 2023, one that finds the third postulate's mechanism in the August 2022 entry, and one that dates the May manuscript to May 8, where claim 15 tests the long manuscript's other date

for it, May 20; the dossier lists those I found. The distribution of verdicts therefore describes this set, not the reliability of my retrospective accounts in general. Part of my later account was drafted within the AI-assisted collaboration described in Section 10.2 and adopted by me. Such passages are my account, since I adopted them, but an error in one has a different source from an error of memory, so Table 8 marks whose wording each claim is in.

Table 8: Later accounts tested against earlier artifacts, claim by claim. LM: the long manuscript, cited from the source of version 519 (PhilArchive, 19 Oct. 2026), a correspondence I state and the archive's version trail does not show. AL: the annotation layer, repository commit 8e54a6e (20 Oct. 2026). ED: an earlier draft of this paper (Sept. 2026). "Mine": in my own wording; "drafted": drafted within the AI-assisted collaboration and adopted by me. Grades: independent (a third party dates the evidence) or self-dated.

| | Later claim (source; wording) | Earlier evidence | Verdict (grade) |
|---|---|---|---|
| *February 8, 2023* | | | |
| 1 | It was "the first axiomatic designation" of the framework (LM; mine) | The January manuscript states "three fundamental rules of reality" and calls them "basic laws," having called its principles "postulates" in a footnote; the February entry quotes them as its "Old Version" | Contradicted (self-dated) |
| 2 | It was "the fundamental moment in which I first used the parent and child Aethic terms" (LM; mine) | The January manuscript defines a child and a parent Ethus; its rule 3, and the entry's own "Old Version," read "parent Ethuses" | Contradicted (self-dated) |
| 3 | It saw "the devising of invalid Aethae for the first time" (LM; mine) | January rule 3 and its gloss already concern invalid states | Contradicted (self-dated) |
| *The January manuscript* | | | |
| 4 | It has "the three rules assembled and numbered for the first time in manuscript" (AL; drafted) | The October 2022 rules are numbered in the journal; no earlier manuscript states numbered rules | Supported (self-dated) |
| 5 | It precedes the February entry by "ten days" (AL, ED; drafted) | 28 Jan. by PDF metadata; 8 Feb. by the journal | Contradicted: eleven (self-dated) |
| 6 | The February entry "boxes these exact sentences" and "redesigns them" (AL; drafted) | The entry quotes the three rules verbatim as its "Old Version" and replaces them | Supported (self-dated) |
| 7 | It boxes them "as 'the old laws' " (AL, LM; drafted) | The box is headed "Old Version"; the entry speaks of "the old second law and third law" | Contradicted as a quotation (self-dated) |
| *The January paradox* | | | |
| 8 | Footnote 6's paradox "is the outstanding question the May form closes" (LM, AL; drafted) | In May the unknown slit stays in superposition as "one slit but not the other in both ways," after the third postulate eliminates "both slits at once" | Supported (self-dated), with a note: the distinction of Section 5.3 does the closing, and the third postulate makes it available |

| | Later claim (source; wording) | Earlier evidence | Verdict (grade) |
|---|---|---|---|
| *The May form and its date* | | | |
| 9 | The first postulate has not changed since May 2023, from "the framework in the axiomatic sense has not changed since May 8, 2023" (LM, AL; drafted) | Stated in its May wording in the October 2023 talk (24:50); restated in new wording in the long manuscript by 2026 | Open: unchanged to Oct. 2023 (independent); the 2026 restatement is not coded here |
| 10 | The second postulate has not changed (the same sentence; elsewhere in LM, "content fixed, dress since evolved"; drafted) | Stated with "statistically independent" through July 2024 and in terms of blanks by April 2025, after the 2024 drafts distinguish the terms | Open (independent): turns on the decomposition reading (see text) |
| 11 | The third postulate has not changed (the same sentence; drafted) | Stated in its May wording in the October 2023 talk (43:26) and, by my reading, version 1; restated by 2026 over "proper" children, in terms of validity | Open: unchanged to Oct. 2023 (independent), and to Nov. 2024 by my reading; the 2026 restatement is not coded here |
| 12 | The final form dates from May 8 (the same sentence; AL: "the two-generation form arrives May 8"; drafted) | The May 7 entry states the candidate and argues the detector case with it; the May 9 entry places "officially cracking the measurement problem" on the previous day; neither records the form reached | Compatible, not discriminating (self-dated): the rival is that May 8 completed the May 7 argument, and the two-generation form arrived by May 23 |
| 13 | The arXiv confirmation of that day dates "my contemporaneous judgment that the work was ready" (LM; drafted) | The message dates an account event; the May 9 entry ties the event to that judgment | Supported (self-dated link to an independently dated event) |
| 14 | The May manuscript's date, May 20, has "the same grading" as the January manuscript's, whose day and second are given "by the file's metadata" (ED; drafted) | The deposited PDF's metadata gives 23 May 2023, 17:10:04 UTC | Contradicted (self-dated), reading "the same grading" as the same basis |
| 15 | The May manuscript is "of May 20, 2023" (LM; drafted) | No artifact is dated May 20; the May 9 entry reports the paper already set in LaTeX; the deposited file was produced on May 23 | Open (self-dated) |
| *The candidates for the third postulate* | | | |
| 16 | "the May page rejecting and refining" a candidate as too strong (ED; drafted) | The May manuscript rejects the one-generation candidate as "not quite accurate"; the May 7 entry had stated it, in substance, as a postulate | Supported (self-dated) |

| | Later claim (source; wording) | Earlier evidence | Verdict (grade) |
|---|---|---|---|
| 17 | A "too-strong/too-weak elimination" is "visible across dated prints," the January form the too-weak candidate (ED; LM similar; drafted) | No text rejects an earlier form as too weak; the January and February forms survive as consequences; my 2026 account calls the final postulate "slightly stronger" | Compatible, not discriminating (self-dated): the rival is that the January form was superseded and kept as a consequence, and only the too-strong candidate was eliminated |
| *The two superpositions* | | | |
| 18 | Both were "present extensionally, insisted to be distinct, months before either had a name" (ED; LM similar; drafted) | On the double-slit case: May 2023 (self-dated) and the October 2023 talk (independent); the names first attested in July 2024 | Supported (independent, from Oct. 2023), with a note: "extensionally" holds for the double-slit cases; the general definition is the extension of Section 5.3 |
| 19 | "content before name" is "the record's recurring signature" (ED; drafted) | One naming is coded here | Open (independent) |
| *The 2022 entries* | | | |
| 20 | "statistical independence" in the August 2022 entry was "simply a semantic convention" in place of blank (LM; mine) | August 2022: the unknown set dropped for independence; June 2024: "not statistically independent ... but it's still blank" | Open (self-dated): turns on the decomposition reading (see text) |
| 21 | The push for independence moved blankness away from "a particular pin being known or not in isolation" (LM; mine) | The entry drops a design of "a set of knowns and then a set of unknowns" for independence from the Ethus, "over the past few days" | Supported (self-dated) |
| 22 | It moved blankness to "indistinguishability across an entire attribute of pins" (LM; mine) | The entry speaks of "things" and of a "concept"; the attribute as unit appears in the October 2022 rules: "An Attribute may have an Anti-Ethus component and an Ethus component" | Compatible, not discriminating (self-dated): the rival is that the entry's unit is the thing or concept, and the attribute arrives in October 2022 |
| 23 | The August 2022 entry "already carries the ordering" of probability first, the wavefunction its summary (LM; drafted) | "the actual probability density function is very much in my Ethus, and this itself is the observed quantum wave function" | Compatible, not discriminating (self-dated): the sentence also fits a possible conflation of the wavefunction with its associated density $\|\psi\|^2$ |

Claims 10 and 20 need the sense-by-use rule, and they show its limits. Two readings of the

2022 and 2023 term "statistically independent" fit the texts. On one, the term was looser than its textbook sense: the January manuscript calls the Anti-Ethus "statistically independent to the Ethus" and, in the next paragraph, lets the Ethus hold "that P(y $<$ 0) $>$ 70%" for an Anti-Ethus variable $y$. On the other, the texts decompose such a variable: the January manuscript divides any property "partially correlated to the Ethus" into "an Ethus component and an Anti-Ethus component," and the October 2023 talk puts the skew in the state and leaves outside it "a further subcomponent of the property describing the random quantile on that distribution" (23:15), so that the term keeps its textbook sense for what remains. The 2024 drafts then separate the later pair of terms: an attribute about which a little is known is, in the speaker's words, "not statistically independent ... but it's still blank" (June 12, 2024, 19:38). The postulate is still stated with "statistically independent" in July 2024 and is described in terms of blanks by April 2025 ("with the blanks go to everything," 45:22 of the talk of April 27, 2025). Whether the later postulate restates the earlier one turns on the decomposition reading, which the record poses sharply without settling, so both claims are open; claim 20's "simply a semantic convention" also passes over the 2024 distinction. Claims 21 and 22 concern an earlier step, the replacement in August 2022 of an "unknown set" by a relation of independence, which the entry records within days of the change.

The first three claims are small errors, but they have the shape the protocol exists to catch. In my own history, a designation and two firsts moved to the entry that rephrased the rules, eleven days after a manuscript that already contains them; both dates are self-reported, but the February entry quotes the January rules, so the order is fixed within the record. The entry itself suggests how the move happened. Its remark that the new second law's contrapositive "is also precisely how we initiate the concept of invalid Ethuses, which the third law after that only runs with" concerns the order of the new laws, and read later it can pass for a chronology.

Of the twenty-three claims, the record contradicts six, supports seven, fits four without discriminating, and leaves six open. Three of the six contradictions are misdated firsts in my own wording; the other three, an arithmetic slip, a misquotation, and a misread file date, are in drafted text that I adopted. Two supported claims needed a note rather than a correction: claim 8 is right that the May form closes the January paradox, and its note says how, and claim 18's "extensionally" holds for the double-slit cases and not for the general definition. Claims 9 and 11 are open for a reason of their own: the first and third postulates keep their May wording in the October 2023 talk, and by 2026 both have been restated, a transition this paper does not code. None of these verdicts touches the framework's content. They concern what its author later said about when and how that content arose, which is the question the record can answer.

## 7 The two-stage hypothesis, made codable

**Hypothesis: two stages**

Inference-driven development tends to move from formulations that rely on agent-centred analogy and familiar imagery and leave their constraints implicit, to formulations that state their constraints explicitly and remain stable across later versions; and the move is concentrated at a few transitions rather than spread evenly across the record.

I first stated this as a "charming" stage, in which a principle is intuited from a subtly anthropocentric vantage, giving way to a "structural" one. "Charming" cannot be coded without hindsight, so the hypothesis is restated with four markers, scored on the statement of a formulation

rather than on the prose around it: (M1) reliance on agent-centred analogy, present or absent; (M2) dependence on familiar imagery, present or absent; (M3) explicitness of the constraint, where 0 means it appears only in a worked case, 1 a general statement whose key terms are fixed only by examples, and 2 a general statement whose key terms are defined in the text or in an earlier text it cites; and (M4) stability, whether the formulation recurs as a stated principle, in its wording or quantifier structure, in later versions, recurrence as a derived consequence not counting. M3 scores definability, not logical complexity. M4 is stability, not invariance under redescription, which would need equivalent descriptions of one situation, such as the January manuscript's doors and slits, compared for the verdicts a formulation gives them; that test is not run here. On a track chosen to end at a surviving form, M4 is fixed for the last formulation by the choice of track and carries no evidence there. Two historical cases suggested the hypothesis to me, Feynman's lecture "There's Plenty of Room at the Bottom" [40] and Einstein's route from the equivalence principle to general covariance. They are its source, not its evidence.

Three alternatives predict much the same surface. On ordinary learning, constraints become explicit gradually and formulations stabilize when they stop failing. On formalization, explicitness and stability rise while content stays fixed. On retrospective organization, later accounts impose the two-stage shape on records that do not show it. On a coded record the alternatives lead one to expect different patterns: ordinary learning, markers moving at different times; formalization, M3 rising across transitions coded clarification or renaming; retrospective organization, the shape in later accounts but not in the contemporaneous artifacts; the hypothesis, the markers moving together at a transition coded as extension or revision.

Table 9: The markers scored on the third postulate's track.

| Formulation of the condition | Date | M1 | M2 | M3 | M4 |
|---|---|---|---|---|---|
| A future iteration of the Ethus that "can potentially contain a current element of my Anti-Ethus" excludes superposition | 22 Aug. 2022 | present | absent | 2 | No |
| Rule #1, "No contradictions allowed!", applied to a superposition: "that rule-breaking-portion of that alternative Ethus must not exist" | 23 Oct. 2022 | absent | absent | 1 | Yes: restated in Jan. 2023 as a "rule of reality" |
| Rule 3: invalid if a contradiction is triggered in the superposition of a parent | Jan. 2023 | absent | absent | 2 | No: a consequence from May, on I1 |
| Law 3: the children of an invalid state are invalid | 8 Feb. 2023 | absent | absent | 2 | No: a consequence from May |
| Postulate 3: the $\exists\forall$ condition over two generations | May 2023 | absent | absent | 1 | Yes, by the choice of track |

Scored on the third postulate's track (Table 9), the record does not show two stages. It shows five formulations in nine months, and the markers do not move together. The agent leaves the statements of the condition by October 2022; imagery is absent from all of them, though the prose around them, May's included, still runs on agents and scenes ("your own Aethus at this moment"; "all hope was not yet lost"). Explicitness does not rise steadily. The August 2022 condition is stated in terms the same paragraph defines ("The Anti-Ethus is simple: it is the set of all possible things which are Rebetary"), the October rule in terms fixed by examples, the January form in defined terms and the February form in terms the January manuscript defines, which the entry quotes, and

the May postulate in a term, "contradict," that the May text fixes by example; January defines it, but May does not cite January. M4 is Yes for the May form, as the choice of track guarantees, and for the October rule, whose prohibition January restates as "one final important rule of reality"; the formulations between recur only as consequences.

The motive for the step to a time-independent condition is recorded twice. The October 2023 talk gives it in retrospect: the speaker "wasn't expecting" the framework's state "to talk to itself from the future because that felt kind of arbitrary," and wanted "a way to crack this code that was completely independent from the flow of time" (39:30). A year earlier, the October 2022 entry had asked "So… are we communicating with the future???" and answered that if so, "it would not feel very fundamental / logically symmetric to me, so I want to find a simpler alternative to this question." The contemporaneous entry supports the talk's account of the motive, and shows the motive already at work a step earlier, in the October 2022 move to a separate Ethus, which became January's parent Ethuses.

Qualitatively, the track looks more like ordinary learning, with a persistent motive and markers moving at different times, than like a concentrated two-stage shift: there were more than two meaningful stages, one transition is a clarification on stated assumptions, and several constructions were abandoned on the way. That comparison is a judgment, not a measurement; a systematic comparison would need each alternative's expectations stated as scores in advance, which this paper does not attempt. The hypothesis's clause on stability cannot be tested on this track at all, since a track that ends at a surviving form is stable at its end by construction; a test needs tracks chosen without regard to survival, including tracks that end in abandonment.

The record contains genuine surprise — the 1:42 PM continuation of the August entry opens "Aw crap. I just thought a little bit more about this thought experiment, and it actually gets weirder still" — but surprise does not protect a two-stage reading from retrospective organization, since the episodes in which surprise is found are selected and classified in retrospect, by me. A genuine test needs a prediction registered before the transition it concerns, coded by rules fixed in advance, on records whose authors do not control the outcome: records other than mine.

# 8 Two hazards for the coder

## 8.1 Installed and independently constrained structure

**Coding distinction: installed and independently constrained structure**

A posit is *installed* when its only evidential role is introduced after an anomaly, to absorb that anomaly's residual, and it adds no independently exposed consequence. A posit is *independently constrained* when the constraints from which it follows were fixed before, or independently of, the anomaly it addresses, or when it adds consequences exposed to test beyond that anomaly.

I first stated this distinction as a slogan: "Failed theories tend to add new substance to the universe. Successful theories tend to add new reasoning to the universe." The slogan invites retrospective sorting, on which every successful posit becomes reasoning by being called entailed and every failed one substance by being called installed. The temporal form above can be applied to dated records before outcomes are known, and it is offered here as a candidate coding distinction requiring validation, not as a historical law. Its nearest ancestors are Einstein's distinction between principle theories and constructive theories [41], a taxonomy of forms that makes no prediction

about failure, and Lakatos's distinction between progressive and degenerating programmes [42], which reads a programme's trajectory where this distinction reads a single posit at its introduction.

Four complications keep the coding honest. Independent motivation is not deductive entailment: a posit can be motivated by constraints fixed in advance without following from them. One proposal can introduce entities, reorganize reasoning, and rescue an existing theory at once. A failed ontology can generate constraints that outlive it. And successful development often keeps some components of a proposal while rejecting others. Maxwell's electrodynamics shows all four. In 1865 he presented a theory of the field against action at a distance, citing "[t]he mechanical difficulties ... which are involved in the assumption of particles acting at a distance," and assuming "an æthereal medium filling space and permeating bodies" [43]. The displacement current that completed his equations had entered in 1861–62 through a mechanical model of that medium, which also yielded the speed of light from electrical measurements [44]; the medium was later dropped and the equations kept. By the slogan, the ether is installed substance on the failed side. By the temporal form, the medium of 1861–62 was a posit with consequences exposed far beyond the anomalies it addressed, and the coding has to record both the posit's failure and the survival of what it generated. Mixed cases of this kind refine the distinction; they are not to be resolved in its favor.

Applied to the record's constructs from their introducing texts, the distinction is harder to run than its statement suggests. The Anti-Ethus was introduced in August 2022, "over the past few days," to replace an "unknown set" its author found "kind of an overly vague concept," before the human-subject puzzle the entry goes on to pose. It absorbed no anomaly, so it is not installed; but nothing in the record shows it following from constraints fixed in advance, or adding a consequence exposed beyond the cases it was built to describe. Its code is left open, and the case shows that the distinction needs a rule for posits introduced to repair a concept rather than to absorb an anomaly. The January manuscript's "compression postulate," that "[a] superposition may never truly collapse – rather it can only compress," serves in footnote 6 as a rejected repair, which suggests installed; but the word arrives in August 2022 as the journal's preferred name for update, months before the paradox, so footnote 6 put an existing posit to a new use. The May distinction between the two superpositions follows from postulates fixed in the same text, which suggests independently constrained; but those postulates were fitted to the double-slit cases (Section 5.2), and the one consequence they add beyond them, the agreement of observers' records, reproduces a known fact. One code is open and two are contested on the record itself, which is what a candidate distinction should show before validation; an independent coder's disagreements are the data that would validate or retire it.

### 8.2 Masking

The sense-by-use rule answers a hazard I have called *linguistic masking*. I introduced the term for one case, the description of one's own cognition in a vocabulary built around other minds [33]. The hazard for a coder is its general form: a standing vocabulary is used for content it was not built to carry, and the words' public sense and their sense in use come apart without writer or reader noticing. It cuts in both directions in an archive. Early texts can use standard words in nonstandard senses — whether the January manuscript's "statistically independent" is a case is the open question of Section 6 — and later names can import senses the earlier texts lacked, as the 2024 names would if they were read back into the 2023 distinction (Section 5.3). A coder who reads early texts by textbook senses, or by the later names, will code reconstruction as continuity and continuity as reconstruction.

Masking also suggests a hypothesis about reception, stated here and not tested: readers given

a novel thesis whose surface language resembles an established position, with the difference stated in the text, will tend to assimilate it to the nearest established category. A test must separate misunderstanding from disagreement, counting only paraphrases that an independent judge finds inaccurate, whatever the judge thinks of the thesis. One instance occurred while this paper was drafted, when an AI reviewer read the conditional "if I happen to be right about my stuff" as governing only a downstream clause, and took an evaluative claim for an evidential one; preserved in the session record, it illustrates the distinction without testing the hypothesis.

A further mechanism, by which a fierce two-sided debate entrenches what both sides share, belongs to a separate comparative study, which would need cases selected by a rule fixed in advance, cases in which the mechanism should be absent, and a specified competing explanation; a position can stay unoccupied because it is camouflaged, but also because it is unproductive, unavailable on the evidence of its day, or not distinct from positions already held [45].

## 9 Judgment at the points of revision

**The art analogy, bounded**

The record shows representations revised by judgments of form and by checks against cases, criteria that were recorded but not codified. Calling such revision an art, guided by judgment and by tips that admit exceptions, describes this recorded practice. It is not a claim that the practice is better described so than procedurally, nor that no procedure could have produced the revisions.

Four targets are easily confused when inductive practice is called an art rather than an algorithm: a universal procedure guaranteed to produce discoveries; a computational model of some discovery processes; normative guidance for assessing inductive moves; and a historical explanation of a particular discovery. The analogy denies only that the first is on offer, and claims nothing against computational models: Simon argued that scientific discovery has a logic [46], programs built in that spirit rediscovered Kepler's third law from data [47], and Schmidhuber's Gödel machines rewrite any part of their own code once they have proved the rewrite useful [48]. An argument from regress — axiomatize inductive practice, and you write the era's assumptions into the axioms — shows only that a method whose fixed axioms include substantive assumptions about the world cannot revise them; a fixed procedure need not contain fixed beliefs about the world. The analogy's prescriptions are tips of the kind Vonnegut gave for writing fiction, which good writers are free to break [49].

The analogy agrees with Norton's material theory on the negative: there are no universal inductive schemas, and "each domain has an inductive logic native to it" [29]. It concerns a stage the material theory is not built to describe, the choice of which representation to revise when the facts that would warrant inferences within it are themselves under revision, and on that stage the record offers process evidence rather than a verdict. At three points of revision it records the criteria its author used: law-likeness, the mutual exclusivity of the laws' contents, and what the laws make visible on February 8, 2023; the rejection of two repairs "for their lack of simplistic clarity" in January; and in May the test of a candidate against the case it had to handle, "close to the final realization, but still not quite accurate." The first two are judgments of form; the third is a check against a case. Whether such criteria are themselves governed by material facts of the domain is the open question between the two views. The protocol of Section 4 is the complement the analysis supports: no rules of discovery in the guaranteed sense, and exacting rules of checking.

In the image I have used before, the outlaw and the sheriff work different shifts on the same job.

# 10 Limits, methods, and what follows

## 10.1 Limits

The record has one subject, who selected it, coded it, and wrote this paper. Its private leg is dated by its author. Its transitions were chosen to exercise the codes, with their outcomes known, and its later accounts were selected by the same person (Section 6). It can show how one framework was built and test what its builder later said about that; it cannot show whether the framework is true, and by itself it cannot support any generalization about discovery.

The comparison class has a selection problem of its own, and it bears on this paper's method: whose records survive to be audited is a fact about access. The notebook studies this paper sets itself beside are studies of celebrated individuals, and any appeal to the historical record of "great individuals" also carries an obligation stated plainly: that record is severely compromised by the systematic exclusion of women, people of color, and other marginalized groups from education and scientific institutions — an exclusion that speaks only to the cruelty of the structures enforcing it and says nothing whatsoever about the capacities of those it targeted, the conflation of denied access with diminished ability being among the most damaging errors in the history of ideas. The irony is particularly bitter[2] when you also consider how the malice behind historical systemic discrimination muzzles the very class of ontology-capable individuals (people able to re-identify what a theory's basic quantities are about) we should be prizing — while the very system doing the muzzling insists that individuals are special. A protocol like this one can be applied only to records that were kept, preserved, and made public, and it inherits the selection that decided which records exist.

## 10.2 Provenance and AI assistance

The 2026 versions of this paper were drafted within an AI-assisted collaboration. Claude models (Anthropic) drafted connective prose and the restructuring of the analysis, and checked citations, quotations, and computations, working from my manuscripts, journal entries, and instructions; the first person, the recollections, and the decisions about what to claim are mine. The same collaboration drafted parts of the later accounts tested in Section 6, which Table 8 marks. The paper's provenance records have four statuses.

**Completed.** The deposits listed in Table 2; the public version trail of the companion manuscript; a public ledger of the drafting sessions and a provenance record of the collaboration, in the `provenance` directory of my repository at commit 8e54a6e [30]; and the supplementary dossier, which freezes every passage this paper quotes or tests, with its location and the SHA-256 hash of its source file.

**Planned.** Deposit of the full session record, the conversation exports with their SHA-256 hashes, which will establish the integrity of those bytes from the date of deposit, not their earlier existence; and deposit of the earlier draft of this paper, identified now by its hash in the dossier.

[2]This disconnect is one of those rare things that actually brings me physical rage: the juxtaposition of denying individuals gratitude because of race or gender with then targeting those same people for reeling under that discrimination — for struggling to reproduce the output stats of the white men those same institutions babied.

**AI-assisted checking.** Pre-submission reviews by language models, of the drafting model's family and of another family.

**Independent human review.** None yet; it precedes submission.

The logs record who contributed what to the text. They do not by themselves settle questions of intellectual authorship, which turn on what the records show about where ideas originated, the question Sections 5 and 6 take up for the framework's postulates.

### 10.3 What follows

The analysis supports five practices, stated as practices rather than rules. Keep dated records while the work is under way, and anchor their hashes in an independent timestamp when they are written; this record's private leg was not anchored, and no later deposit can repair that. Separate what each artifact can establish before using it as evidence (Table 1). Code transitions component by component, from the artifacts, before consulting what survived, and state the interpretive assumptions each code needs. Test later accounts against earlier artifacts, one's own accounts first, claim by claim, and mark whose wording each is in; the errors found in Section 6 would otherwise have stood in my own history. And keep abandoned constructions on the page, since a record pruned to its successes hides exactly the revisions that distinguish development from reconstruction. The single-subject limitation will dissolve only as other builders keep, and open, records of their own.

**A note on sources.** The journal entries quoted here are from the public export of the journal [30]; most are also reproduced in the long manuscript [1]. The 2023 manuscripts are quoted from the deposited PDFs [31, 32], and the summary of October 2023 from its hosted export [34]; the talks are quoted from the transcripts described in Section 3, with timestamps for checking against the audio [35]. The supplementary dossier gives, for every quoted or tested passage, the artifact, version, and location, and for every source file its SHA-256 hash. Recollections are mine, at most with drafted wording, and will be checkable against the session record once it is deposited.

## A Coding sheet

For each transition, a coder records:

1. the construct, identified by its function in the framework;
2. the earlier formulation: passage, artifact, version, date, and dating basis (Table 1);
3. the later formulation, with the same fields;
4. the components of each formulation, each a claim located in its text, with senses fixed by use (Table 4);
5. for each component, a code from Table 3, the passage in the other formulation that supplies it, and any interpretive assumption the code needs;
6. the transition's code, derived from item 5 alone, and, where an assumption is needed, the code the transition receives if the assumption is denied;
7. the strongest competing reading, stated as the denial of an assumption in item 5 or as a rival code for a named component, and the evidence that would decide between it and the code;

8. the construct's survival, recorded after item 6.

For each later account, the coder splits the account into atomic claims, each asserting one thing, and records for each its source, version, location, and whose wording it is; the earlier artifacts consulted; and a verdict from Table 3, with the grade of its best evidence. Where the coder's entries differ from those of Tables 5–8, the difference and its ground are the data the recoding exists to produce.

The primary texts are in my repository at commit 8e54a6e [30]: the January and May 2023 manuscripts (`history/Jan2023Aethic.pdf`, `history/Original_Aethic.pdf`); the journal export (`development/journals/`, file `Foundational Journal Entries.md`); the summary of October 2023 (`history/MajorRulesoftheAethus.html`); and the verbatim restoration, with transcripts of the talks (`history/marked_original.html`). The preprint's version trail is at `https://philarchive.org/versions/BENARA-5`. The supplementary dossier freezes every passage used here, with its location and the SHA-256 hash of its source file.

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